

Magnetism and Electro Magnetism

1. Magnet has _____.

- a) two poles
- b) one pole
- c) no poles
- d) multiple poles

Correct answer: a) two poles

2. Like magnetic poles _____.

- a) attract
- b) repel
- c) neutral
- d) cancel

Correct answer: b) repel

3. Earth's magnetic field: north pole points to _____.

- a) geographic north
- b) magnetic south (geographic north)
- c) equator
- d) south

Correct answer: b) magnetic south (geographic north)

4. Magnetic meridian is plane containing _____.

- a) magnetic axis and vertical
- b) geographic north
- c) both
- d) none

Correct answer: a) magnetic axis and vertical

5. Angle of dip is angle between magnetic field and _____.

- a) horizontal
- b) vertical
- c) equator

d) pole

Correct answer: a) horizontal

6. Magnetic field lines are _____.

a) closed curves

b) open

c) straight

d) random

Correct answer: a) closed curves

7. Direction of magnetic field: _____.

a) tangent to field line

b) perpendicular

c) radial

d) none

Correct answer: a) tangent to field line

8. Crowding of field lines means _____.

a) strong field

b) weak field

c) no field

d) uniform

Correct answer: a) strong field

9. Unit of magnetic field strength _____.

a) tesla

b) gauss

c) weber

d) ampere

Correct answer: a) tesla

10. 1 tesla = _____ gauss.

a) 10^4

b) 10^3

c) 10^2

d) 10

Correct answer: a) 10^4

11. Magnetic field due to straight conductor: _____.

a) right hand thumb rule

b) left hand

c) Fleming left

d) right

Correct answer: a) right hand thumb rule

12. Field inside solenoid is _____.

a) uniform

b) non-uniform

c) zero

d) radial

Correct answer: a) uniform

13. Solenoid field strength increases with _____.

a) more turns

b) soft iron core

c) both

d) less current

Correct answer: c) both

14. Electromagnet is _____.

a) temporary magnet

b) permanent

c) natural

d) artificial

Correct answer: a) temporary magnet

15. Fleming's left hand rule for _____.

a) motor

b) generator

c) solenoid

d) bar magnet

Correct answer: a) motor

16. Fleming's right hand rule for _____.

a) motor

b) generator

c) force

d) current

Correct answer: b) generator

17. Electric motor converts _____ to _____.

a) electrical to mechanical

b) mechanical to electrical

c) heat to mechanical

d) chemical to electrical

Correct answer: a) electrical to mechanical

18. DC motor has _____ commutator.

a) split ring

b) slip ring

c) both

d) none

Correct answer: a) split ring

19. AC generator has _____.

a) slip ring

b) split ring

c) brushes only

d) armature only

Correct answer: a) slip ring

20. Generator works on _____.

a) electromagnetic induction

b) motor effect

c) magnetic effect

d) heating

Correct answer: a) electromagnetic induction

21. Faraday's law: $\text{emf} \propto$ _____.

a) rate of change of flux

b) flux

c) current

d) resistance

Correct answer: a) rate of change of flux

22. Lenz's law: induced current opposes _____.

a) change in flux

b) flux

c) motion

d) all

Correct answer: a) change in flux

23. Transformer works on _____.

a) AC

b) DC

c) both

d) none

Correct answer: a) AC

24. Step-up transformer: _____.

a) increases voltage

b) decreases voltage

c) same

d) current increases

Correct answer: a) increases voltage

25. Turns ratio $N_p/N_s < 1$ for _____.

a) step-up

b) step-down

c) same

d) ideal

Correct answer: b) step-down

26. Eddy currents cause _____.

a) energy loss as heat

b) increase efficiency

c) no effect

d) cooling

Correct answer: a) energy loss as heat

27. Laminated core reduces _____.

a) eddy currents

b) hysteresis

c) both

d) none

Correct answer: a) eddy currents

28. Self induction opposes change in _____.

a) current

b) voltage

c) flux

d) power

Correct answer: a) current

29. Mutual induction between _____.

a) two coils

b) single coil

c) solenoid

d) bar magnet

Correct answer: a) two coils

30. Earth's magnetism due to _____.

a) molten iron core

b) external field

c) sun

d) moon

Correct answer: a) molten iron core

31. Magnetic elements: declination and _____.

a) dip

b) inclination

c) both

d) none

Correct answer: c) both

32. Declination zero at _____.

a) magnetic equator

b) agonic line

c) poles

d) tropics

Correct answer: b) agonic line

33. Dip 90° at _____.

a) magnetic poles

b) equator

c) tropics

d) agonic

Correct answer: a) magnetic poles

34. Bar magnet field lines emerge from _____.

a) north pole

b) south pole

c) equator

d) centre

Correct answer: a) north pole

35. Neutral point for north pole of magnet facing geographic north is _____.

a) broadside-on

b) axial

c) equatorial

d) end-on

Correct answer: a) broadside-on

36. Force on conductor in magnetic field: $F =$ _____.

a) $BIL \sin\theta$

b) BI/L

c) BL/I

d) zero

Correct answer: a) $BIL \sin\theta$

37. Motor effect discovered by _____.

a) Oersted

b) Faraday

c) Ampere

d) Coulomb

Correct answer: a) Oersted

38. Current carrying conductor in magnetic field experiences force _____.

a) perpendicular to B and I

b) parallel

c) same direction

d) no force

Correct answer: a) perpendicular to B and I

39. Right hand thumb rule: thumb current, fingers _____.

a) B direction

b) force

c) both

d) none

Correct answer: a) B direction

40. In DC motor, split ring commutator ensures _____.

a) unidirectional torque

b) bidirectional

c) no torque

d) constant speed

Correct answer: a) unidirectional torque

41. AC generator produces _____.

a) sinusoidal emf

b) DC

c) pulsating

d) constant

Correct answer: a) sinusoidal emf

42. Slip rings provide _____.

a) continuous connection

b) split

c) commutator

d) brushes

Correct answer: a) continuous connection

43. Induced emf maximum when flux change _____.

a) maximum rate

b) minimum

c) zero

d) constant

Correct answer: a) maximum rate

44. Transformer efficiency nearly _____.

a) 100%

b) 50%

c) 90%

d) 75%

Correct answer: a) 100%

45. Core type transformer has _____.

a) laminated core

b) air core

c) iron core solid

d) copper core

Correct answer: a) laminated core

46. Hysteresis loss reduced by _____.

a) soft iron

b) hard iron

c) steel

d) aluminium

Correct answer: a) soft iron

47. Induction furnace uses _____.

a) eddy currents

b) motor

c) generator

d) transformer

Correct answer: a) eddy currents

48. Dead beat galvanometer has _____.

a) high damping

b) low damping

c) no damping

d) variable

Correct answer: a) high damping

49. Vibration magnetometer measures _____.

a) magnetic moment

b) field

c) both

d) current

Correct answer: c) both

50. Tan A position: magnet _____.

a) broadside-on

b) end-on

c) equatorial

d) axial

Correct answer: a) broadside-on

51. Time period $T = 2\pi \sqrt{I/M Bh}$ for _____.

a) vibration magnetometer

b) deflection

c) $\tan A$

d) deflection magnetometer

Correct answer: a) vibration magnetometer

52. $Bh = \mu_0 M / (2\pi d^3)$ for _____.

a) axial position

b) equatorial

c) broadside

d) pole

Correct answer: a) axial position

53. $Bh = \mu_0 M / (4\pi d^3)$ for _____.

a) equatorial

b) axial

c) end-on

d) broadside

Correct answer: a) equatorial

54. Permanent magnets made of _____.

a) steel

b) soft iron

c) aluminium

d) copper

Correct answer: a) steel

55. Electromagnets use _____.

a) soft iron

b) steel

c) both

d) none

Correct answer: a) soft iron

56. Retentivity high in _____.

a) permanent magnets

b) electromagnets

c) soft iron

d) air

Correct answer: a) permanent magnets

57. Coercivity low in _____.

a) soft iron

b) hard steel

c) alnico

d) ferrite

Correct answer: a) soft iron

58. Demagnetisation by _____.

a) hammering

b) heating

c) both

d) AC coil

Correct answer: c) both

59. Alnico is alloy of _____.

a) Al, Ni, Co

b) Fe, C

c) Cu, Zn

d) Ag, Au

Correct answer: a) Al, Ni, Co

60. Ferrites used in _____.

a) high frequency

b) low frequency

c) DC

d) motors

Correct answer: a) high frequency

61. Cyclotron accelerates _____.

a) charged particles

b) neutral

c) electrons only

d) protons only

Correct answer: a) charged particles

62. Frequency in cyclotron $f =$ _____.

a) $qB/(2\pi m)$

b) B/qm

c) m/qB

d) constant

Correct answer: a) $qB/(2\pi m)$

63. Magnetic flux $\Phi =$ _____.

a) $BA \cos\theta$

b) B/A

c) A/B

d) zero

Correct answer: a) $BA \cos\theta$

64. Motional emf = _____.

a) $B l v$

b) $B v / l$

c) $l v / B$

d) zero

Correct answer: a) $B l v$

65. In generator, speed increases emf _____.

a) increases

b) decreases

c) constant

d) zero

Correct answer: a) increases

66. Self inductance L : $\Phi =$ _____.

a) $L I$

b) I/L

c) L/I

d) constant

Correct answer: a) $L I$

67. Unit of inductance _____.

a) henry

b) weber

c) tesla

d) ampere

Correct answer: a) henry

68. Mutual inductance M : $\text{emf} = -M \, dI/dt$ for _____.

a) secondary

b) primary

c) both

d) none

Correct answer: a) secondary

69. Transformer: $V_p/V_s =$ _____.

a) N_p/N_s

b) N_s/N_p

c) I_p/I_s

d) all

Correct answer: a) N_p/N_s

70. Step down: N_p _____ N_s .

a) $<$

b) $>$

c) $=$

d) none

Correct answer: a) <

71. Efficiency $\eta = (\text{power output} / \text{input}) \times \underline{\hspace{2cm}}$.

a) 100%

b) 50%

c) 90%

d) variable

Correct answer: a) 100%

72. Losses in transformer: copper, iron, .

a) hysteresis, eddy

b) mechanical

c) both

d) none

Correct answer: a) hysteresis, eddy

73. Auto transformer has .

a) single winding

b) two windings

c) three

d) none

Correct answer: a) single winding

74. Choke coil has .

a) high L, high R

b) high L, low R

c) low L, high R

d) low both

Correct answer: b) high L, low R

75. Inverter converts .

a) DC to AC

b) AC to DC

c) AC to AC

d) DC to DC

Correct answer: a) DC to AC

76. Rectifier: AC to _____.

a) DC

b) AC

c) both

d) none

Correct answer: a) DC

77. Domestic induction cooker uses _____.

a) eddy currents

b) motor

c) generator

d) transformer

Correct answer: a) eddy currents

78. MRI uses _____.

a) strong magnets

b) weak

c) no magnets

d) electromagnets only

Correct answer: a) strong magnets

79. Maglev trains use _____.

a) magnetic levitation

b) friction

c) wheels

d) air cushion

Correct answer: a) magnetic levitation

80. Compass needle aligns with _____.

a) magnetic meridian

b) geographic

c) both

d) random

Correct answer: a) magnetic meridian

81. Variation in declination over time _____.

a) secular variation

b) daily

c) annual

d) none

Correct answer: a) secular variation

82. Isogonic lines: _____.

a) same declination

b) same dip

c) same field

d) poles

Correct answer: a) same declination

83. Isoclinic lines: _____.

a) same dip

b) declination

c) field

d) horizontal component

Correct answer: a) same dip

84. $B_h = H \cos \delta$, $B_v = H$ _____.

a) $\sin \delta$

b) $\cos \delta$

c) $\tan \delta$

d) $\cot \delta$

Correct answer: a) $\sin \delta$

85. Total field $B =$ _____.

a) $\mu_0 H$

b) H / μ_0

c) H

d) zero

Correct answer: a) $\mu_0 H$

86. Bar magnet magnetic moment $M = \text{pole strength} \times \underline{\hspace{2cm}}$.

a) magnetic length

b) physical length

c) $2l$

d) l

Correct answer: a) magnetic length

87. Torque on bar magnet $\tau = \underline{\hspace{2cm}}$.

a) $M B \sin\theta$

b) M/B

c) B/M

d) zero

Correct answer: a) $M B \sin\theta$

88. In uniform field, bar magnet in $\underline{\hspace{2cm}}$.

a) stable equilibrium along field

b) unstable broadside

c) both

d) none

Correct answer: c) both

89. Deflection magnetometer: $\tan A$ and $\underline{\hspace{2cm}}$.

a) $\tan C$

b) vibration

c) both

d) none

Correct answer: c) both

90. In $\tan A: B = (\mu_0/4\pi) (M / d^3) \underline{\hspace{2cm}}$.

a) $\tan\theta$

b) $\times \tan\theta$

c) $/ \tan\theta$

d) $\sin\theta$

Correct answer: b) $\times \tan\theta$

91. Pole strength $q_m =$ _____.

a) $M / 2l$

b) $2l / M$

c) $M l$

d) l / M

Correct answer: a) $M / 2l$

92. Natural magnet _____.

a) lodestone

b) steel

c) soft iron

d) alnico

Correct answer: a) lodestone

93. Artificial permanent: _____.

a) alnico, ferrite

b) electromagnet

c) both

d) temporary

Correct answer: a) alnico, ferrite

94. Demagnetising field in _____.

a) transverse

b) longitudinal

c) both

d) none

Correct answer: a) transverse

95. Hard magnetic materials have _____.

a) high coercivity

b) low

c) zero

d) variable

Correct answer: a) high coercivity

96. Soft iron: low coercivity, low _____.

a) retentivity

b) high retentivity

c) coercivity

d) both low

Correct answer: d) both low

97. Hysteresis loop area represents _____.

a) energy loss

b) efficiency

c) power

d) current

Correct answer: a) energy loss

98. To magnetise steel, use _____.

a) DC current

b) AC

c) hammering

d) heating

Correct answer: a) DC current

99. Stroking method uses _____.

a) permanent magnet

b) electromagnet

c) both

d) none

Correct answer: c) both

100. Electrical method: solenoid with _____.

a) steel piece inside

b) air core

c) copper

d) aluminium

Correct answer: a) steel piece inside